

KHOLOUD KHALED AHMED

Artificial Intelligence Student

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CAREER OBJECTIVE

Third-year Artificial Intelligence student with hands-on experience building and evaluating machine learning models across classification, regression, and optimization problems. Skilled in data preprocessing, model evaluation, and translating analytical results into measurable, data-driven decisions — seeking to apply these skills to real-world AI challenges.

EDUCATION

Faculty of Computers and Artificial Intelligence — Assiut National University

2023 – 2027

B.Sc. in Artificial Intelligence

Expected Graduation: 2027

TECHNICAL PROJECTS

SafeLend: Loan Default Prediction Model (NTI Project) — (2026)

- Built and evaluated classification models (Logistic Regression, Random Forest, XGBoost) on a 32,581-loan dataset with a 21.8% historical default rate across 12 features.
- Cleaned the dataset (removing 165 duplicate and 7 invalid outlier rows) and handled class imbalance through preprocessing, including log transformations.
- Tuned XGBoost with 3-fold cross-validated GridSearchCV, reaching 93% accuracy and an 83.4% F1-score — correctly flagging 1,129 of 1,418 real defaulters (79.6% recall) before loan approval.

Social Media Ad Budget Optimization using PSO — (2026)

- Implemented a binary Particle Swarm Optimization (PSO) algorithm to maximize Return on Investment (ROI) for digital marketing campaigns across 6 social platforms.
- Formulated mathematical models for particle velocity and position updates to automate budget distribution without exceeding the \$10,000 capacity.
- Achieved an optimal selection of 4 platforms, yielding a total ROI of 290 while saving \$3,500 (35%) of the allocated budget.

Car Price Prediction (University Team Project) — (2025)

- Collaborated within a 6-member team to develop and evaluate regression models (Linear, Lasso, Polynomial, KNN) to predict vehicle prices based on multiple categorical and numerical features.
- Executed comprehensive data preprocessing, including IQR outlier removal, dummy variable encoding, and StandardScaler normalization using Pandas and scikit-learn.
- Documented model performance metrics and codebase in a shared repository
- Cleaned a 2,500-record dataset (10→15 features after encoding) with a 90/10 train-test split.

EXPERIENCE & CERTIFICATIONS

Summer Intern | Nile Oil Company (NEEL) — (10-Day Program, July 2026)

- Explored enterprise server architectures and gained practical exposure to Security Operations Center (SOC) programs and corporate network systems.

Machine Learning Summer Training (120 Hours) | NTI & ITIDA — (Aug 2026)

- Completed intensive technical and freelancing training, graduating with an exceptional score of 93.5%.

Python Programming Basics | ITI / Mahara-Tech — (Oct 2025)

- Completed foundational programming training through the Artificial Intelligence Academy.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL fundamentals, MATLAB, GAMS, HTML/CSS.
- **Data Science Libraries:** Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, Plotly.
- **Supervised Machine Learning:** Linear/Lasso/Polynomial Regression, Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, XGBoost.
- **Unsupervised Machine Learning:** Clustering algorithms (e.g., K-Means), Dimensionality Reduction (e.g., PCA).
- **AI & Optimization:** Particle Swarm Optimization (PSO), Algorithmic Complexity.
- **Data Processing & BI:** Data Analysis, Data Visualization, Feature Scaling, Outlier Detection (IQR), Class Imbalance Handling.

Soft Skills

Team Collaboration, Problem-Solving, Analytical Thinking, Time Management, Adaptability, Communication, Attention to Detail, Critical Thinking.